A brief digression on exchange statistics

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Disclaimer: lecture notes may contain "typo" type errors (factors of 2, minus signs, etc...) if something like this looks wrong there is a good chance it is, please let me know

Unfortunately I don't know of a good (or bad for that matter) text book which covers this in the way I do here, but a condensed matter physics textbook would probably be the best place to find background reading

1 Motivation

- Given our recent interest in quantum chemistry, it is worth discussing how to represent electronic systems, and in particular how to do it on a quantum computer
- Useful to get a feeling for what terms like Slater determinant actually mean in terms of the physical electrons, as well as where the Pauli exclusion principle actually comes from
- We can get a sense of what makes the quantum chemistry problems we aim to solve "hard"
- Many important behaviours of photons come from the fact that they are Bosons (the Hong-Ou-Mandel effect is a key example); Boson sampling is hard to simulate classically exactly because of exchange statistics, the outcome of the same experiments with classical billiard balls could be solved by hand

2 Symmetric and anti-symmetric wavefunctions

- Unlike classical particles, quantum particles can be truely indistinguishable, this leads to a concept which fundamentally doesn't exist classically, **exchange statistics**

- Key concept: if particles are really completely identical, than a multi-particle state built out of single particle states should be represented as a superposition of every possible ordering of the particles ($n_{\text{particle}}!$ in total),

$$|\Psi\rangle = \frac{1}{\sqrt{n_{\text{particle}}!}} \sum_{\sigma \in \text{permutations}} \exp(i\phi(\sigma)) |\psi_{\sigma_1}\rangle \otimes |\psi_{\sigma_2}\rangle \otimes ... \otimes |\psi_{\sigma_{n_{\text{particle}}}}\rangle) \tag{1}$$

$$= \frac{1}{\sqrt{n_{\text{particle}}!}} \sum_{\sigma \in \text{permutations}} \exp(i\phi(\sigma)) \bigotimes_{j=1}^{n_{\text{particle}}} |\psi_{\sigma_j}\rangle \tag{2}$$

- The phase $\phi(\sigma)$ means that we can define different kinds of particles based on what phases we give in this superposition, but there are some limitations:

  1. Needs to be symmetric in the sense that re-labelling the particles (which cannot change anything physically) must be equivalent to adding a global phase (which cannot change anything physically). This means that we really only get one phase to play with, the phase we get when we swap adjacent particles, Φ (more about how this works later), two-particle example $\Psi = |\psi_1\rangle \otimes |\psi_2\rangle + \exp(i\Phi)|\psi_2\rangle \otimes |\psi_1\rangle$
  2. In 3 or more dimensions, whether a particle moving in a loop goes "around" another particle is ambiguous, therefore exchanging a pair of particles twice must give a total phase of 0. Outside of 2-dimensional or quasi-1-dimensional condensed matter systems, we can only have $\Phi = 0$ (Bosons) or $\Phi = \pi$ (Fermions)[1]

- Fortunately, relative phases only matter when we return to the same configuration (determine what kind of interference occurs), therefore we can define phases based on how many exchanges between the jth and the $j+1$th particle must happen in order to get from $\otimes_j |\psi_j\rangle$ to a given configuration.

  - If we consider an operation which swaps particle k and l, but leaves all other in the same configuration than this will always require an odd number of adjacent swaps, why? We first can do m operations to get the $|\psi_k\rangle$ and $|\psi_l\rangle$ next to each other, one operation to swap them, and m more operations to return everything to the original positions, leading to $2m+1$ total swaps
  - A systematic way to calculate the phase of each permutation is to write a matrix of the different wavefunctions and take a determinant, for this reason multi-body wavefunctions defined as sum over orderings as in equation 2 are known as **Slater determinants**

[1] There are many exotic behaviours you can get from 2D "anyon" systems, including cases where matrix operations can be encoded in exchanges in a way which supports universal quantum computation, but that isn't relevant to what we are discussing here

- It follows that swapping $|\psi_k\rangle$ and $|\psi_l\rangle$ for any $k \neq l$ only will lead to a global $-$ sign ($\exp(i\pi(2m+1))$ where m is an integer) in the total wavefunction if the particles are Fermions
- If we then define $|\psi_k\rangle = |\psi_l\rangle$ (again for any $k \neq l$), than the two orderings must cancel, this is the **Pauli exclusion principle**, two Fermions cannot occupy the same state
- For Bosons it is fine to define $|\psi_k\rangle = |\psi_l\rangle$, as everything has a positive sign, an unlimited number of Bosons can occupy the same state (note that the equivalent of the determanent for Bosons is called a "**permanent**")
- Equation 2 represents a state with a fixed number of particles, but as we know from quantum optics, many interesting physical systems consist of superpositions of states with a different number of photons, therefore $|\Psi\rangle$ does not define an arbitrary state (similarly not all Fermionic systems can be represented by a single Slater determinant, this is the Hartree-Fock approximation)
- Sums of wavefunctions taking the form in equation 2 can represent arbitrary systems
- What we have done here is all very pedogogical and nice mathematically, but it is not very practical for calculations[2], why?
  - We need to keep track of $n!$ terms for any system involving n particles, even if they are in a relatively "simple" state
  - This is *on top of* keeping track of superpositions of such states, which can also be exponential in number
- As long as we know the statistics, we should be able to track exchange statistics automatically, and only keep track of how many particles are in each "mode"
- Fortunately the expectation value of any physically relevant operator will be the same for any permutation, so we don't need the whole expression to calculate values

3 Bosonic and Fermionic operators

- Let's first think of Bosons, because they are what we already used to from quantum optics
  1. Although we haven't made it very explicit $\hat{a}$ and $\hat{a}^\dagger$ are Bosonic operators

[2] recall that we haven't been doing quantum optics this way even though photons are Bosons

2. Since the phase of all permutations are the same for Bosons, there are no "extra" phases based on what other particles may or may not be present
3. The factor of $\sqrt{n}$ in $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ comes from the fact that any of n photons[3] can be annihilated, it can be directly seen in the following way:
   - Remove $|\psi_n\rangle$[4] from each term in the sum from equation 2
   - Since this wavefunction could appear in any of n places, then the new wavefunction can be written

$$\frac{n}{\sqrt{n!}} \sum_{\sigma \in \text{permutations}} \exp(i\phi(\sigma)) \bigotimes_{j=1}^{n-1} |\psi_{\sigma_j}\rangle = \tag{3}$$

$$\frac{n}{\sqrt{n}\sqrt{(n-1)!}} \sum_{\sigma \in \text{permutations}} \exp(i\phi(\sigma)) \bigotimes_{j=1}^{n-1} |\psi_{\sigma_j}\rangle = \sqrt{n}|n-1\rangle \tag{4}$$

   - A similar trick could be applied by **adding** an additional wavefunction any of $n+1$ places (since it could be added on either end), to get $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$

- Fermions are tricker because we need to keep track of the extra minus sign, but they have the advantage that only one Fermion can be in a mode (Fermionic modes tend to be called "orbitals" for historical reasons)
- We need to choose an order for our orbitals, as before we know it doesn't matter how we order them as long as we use the same one consistently
- Since Fermionic modes can only have 1 or 0 Fermions, it is convenient to use Pauli matrices to represent fermionic operators, the following will be useful for us[5]

$$\hat{\sigma}_j^z = (1_2)^{\otimes j} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes (1_2)^{\otimes n-j} \tag{5}$$

$$\hat{\sigma}_j^+ = (1_2)^{\otimes j} \otimes \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \otimes (1_2)^{\otimes n-j} \tag{6}$$

$$\hat{\sigma}_j^- = (\hat{\sigma}_j^+)^\dagger = (1_2)^{\otimes j} \otimes \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \otimes (1_2)^{\otimes n-j} \tag{7}$$

[3] By the notation used elsewhere in the notes this should be n_{photon} but the subscript has been omitted for brevity

[4] by symmetry you could delete any, but the nth is easier to deal with notationally

[5] I am assuming standard qubit matrix notation where the first row/column corresponds to $|0\rangle$ and the second to $|1\rangle$.

- Analogous to the way we treat Bosons, I define Fermionic creation and annihilation operators

$$\hat{c}_j^\dagger = \Pi_{l=1}^{j-1} \hat{\sigma}_l^z \hat{\sigma}_j^+ \tag{8}$$

$$\hat{c}_j = \Pi_{l=1}^{j-1} \hat{\sigma}_l^z \hat{\sigma}_j^- \tag{9}$$

- Since $(\sigma^z)^2 = 1_2$, then $\hat{c}_k^\dagger \hat{c}_l$ "counts" whether the Fermion has gone past an odd or even number of other Fermions (in our ordering, which can be arbitrarily chosen) and applies an appropriate phase

- Playing with this a bit, we can see that this will give us exactly the same phases as if we explicitly represented the whole Slater determinant[6]!, but it only requires one state vector element to represent a state with n Fermions, rather than $n!$

- The definition given in equations 8 and 9 is known as the **Jordan-Wigner transformation** since the transform Fermions into spin variables

- For (classical) numerical matrix calculations, the Jordan-Wigner transformation works great, but the long string of σ_z terms could be annoying if you wanted to program a simulation onto a near term quantum computer, fancier tricks (the Bravi-Kitaev transform) can be used to make this somewhat more efficient (see https://arxiv.org/pdf/quant-ph/0003137.pdf)

4 Non-interacting Bosons and Fermions

- The Hamiltonian for a non-interacting (and number conserving, which is the interesting case for chemistry) system of Fermions can be written as

$$\hat{H} = \sum_j U_j \hat{c}_j^\dagger \hat{c}_j + \sum_{j,k} J_{j,k} \left(\hat{c}_j^\dagger \hat{c}_k + \hat{c}_k^\dagger \hat{c}_j \right) \tag{10}$$

- The J terms correspond to electrons "hopping" between locations (or ''modes"), and U terms correspond to the potential energy of putting an electron at a given mode. Note that this Hamiltonian conserves electron number.

- Let us look concretely how these terms come about if we assume we are solving a three dimensional Shcrödinger equation with a potential (using a set of orbitals $\psi_j(\vec{x})$), the real space Hamiltonian then becomes

$$H = -\frac{\hbar^2}{2m} \nabla^2 + U(\vec{x}) \tag{11}$$

where $\nabla^2 = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} + \frac{\partial^2}{\partial x_3^2}$ where I have used 1 ,2, and 3 to indicate spatial directions

[6] assuming the arbitrary ordering used here is the same as the one used in the determinant

- In this case we have

$$U_j = \int d^3\vec{x}\psi_j^*(\vec{x})\left(-\frac{\hbar^2}{2m}\nabla^2\psi_j(\vec{x}) + U(\vec{x})\psi_j(\vec{x})\right) \tag{12}$$

  and similarly

$$J_{j,k} = \int d^3\vec{x}\psi_j^*(\vec{x})\left(-\frac{\hbar^2}{2m}\nabla^2\psi_k(\vec{x}) + U(\vec{x})\psi_k(\vec{x})\right) \tag{13}$$

  Both integrals can be calculated easily numerically if not analytically

- Since the electrons do not interact, we can diagonalize the Hamiltonian with a single electron (this will be a $n_{\text{mode}} \times n_{\text{mode}}$ matrix), since the energy of one electron does not depend on the others, such a transformation will also diagonalise the whole Hamiltonian

$$\hat{H}_{\text{diag}} = \sum_j E_j \hat{c}_j^\dagger \hat{c}_j \tag{14}$$

- Energy and other expectations can be found by summing each mode individually, non-interacting electronic systems can easily be solved classically (and often by hand)

- Most traditional (i.e. pre-computer) condensed matter physics[7] and chemistry relies on treating electrons as non-interacting, for example solving the Schrödinger equation with spherical symmetry gives the electron orbitals we learn about in high-school, and semiconductor physics (which is needed to build computers so had to come first) is based on analysing at what modes are available in a crystal and how many electrons have filled these modes

- Even in cases too complicated to solve by hand, there is no combinatorial explosion so classical computers can solve these systems easily

- Based on the results for non-interacting Fermions, one might expect that non-interacting Boson systems are also classically easy to solve, however this intuition is wrong!

- Even performing a beamsplitting operation on a single number state leads to a complicated (and entangled) superposition of modes, more operations lead to a combinatorial explosion, and there is no "nice" trick to clean this up

- Finding the number of photons in a given mode means keeping track of all superposition, which amounts to taking a quantity called the permanent[8]

[7] Superconductivity is a notable exception in that it does require interactions between electrons

[8] The permanent is simply the determinant of a matrix but without the minus signs based on parity, the determinant is efficient to calculate because it is basis invariant and is just the product of the eigenvalues in the diagonal basis of the matrix, the permanent is not invariant under basis changes and therefore requires evaluating factorially many terms in general

of the relevant submatrix of a unitary which gives the desired output (and appropriately normalising)

- Based on being hard classically, some companies (Xanadu for example) have made versions of Boson sampling part of their product offerings
- Since viabrational modes of molecules are Bosons, there may be some applications of Boson sampling to chemistry, see for example https://arxiv.org/abs/1412.8427

5 Interacting Fermions

- Electrons are charged particles, so they naturally have a strong coulomb interaction between them, while it is amazing (IMO) that we can get any reasonable results ignoring this interaction (and all other interactions), there are some cases where electron interactions cannot be ignored
- The Coulomb potential (electromagnetic interaction) for example looks like (ignoring constant prefactors)

$$\hat{H}^{(\text{Coulomb})} = \int d^3\vec{x}_1 \int d^3\vec{x}_2 \frac{\hat{n}(\vec{x}_1)\hat{n}(\vec{x}_2)}{|\vec{x}_1 - \vec{x}_2|} = \tag{15}$$

$$\int d^3\vec{x}_1 \int d^3\vec{x}_2 \frac{\hat{c}^\dagger(\vec{x}_1)\hat{c}(\vec{x}_1)\hat{c}^\dagger(\vec{x}_2)\hat{c}(\vec{x}_2)}{|\vec{x}_1 - \vec{x}_2|} \tag{16}$$

- Often it is convenient to solve in a basis other than the one in which the interaction is diagonal in which case we get off-diagonal terms in the interaction (note that creation operations at a single position aren't actually particularly well defined, I show what the integrals look like in reality a little later, there is also a subtlety about electrons not being able to interact with themselves)

$$\hat{H}^{(\text{int})} = \sum_{j\neq p, k\neq l} V_{j,k,l,p}\hat{c}_j^\dagger \hat{c}_k \hat{c}_l^\dagger \hat{c}_p \tag{17}$$

- Why the $\neq$? because an electron cannot interact with itself, a cleaner way to deal with this is to instead define [9]

$$\hat{H}^{(\text{int})} = -\sum_{j,k,l,p} V_{j,k,l,p}\hat{c}_j^\dagger \hat{c}_l^\dagger \hat{c}_k \hat{c}_p \tag{18}$$

- Because $\hat{c}_j^\dagger \hat{c}_j^\dagger = \hat{c}_j \hat{c}_j = 0$, than this expression will automatically evaluate to zero if $j = l$ or $k = p$, automatically eliminating self-interactions

[9] note that for $k \neq l$ $\hat{c}_l^\dagger \hat{c}_k = -\hat{c}_k \hat{c}_l^\dagger$, that is feriomionic operators on different modes **anti**-commute, leading to an extra minus sign

- $V_{j,k,l,p}$ is a tensor whose elements must be calculated in the appropriate basis (lots of inner products, either sums or integrals depended on discrete or continuous), but not computationally hard

- As a concrete example, lets say we have a set of orbitals which are continuous mutually orthogonal and normalised functions $\psi_j(\vec{x})$, where $\vec{x}$ is the spatial coordinate in three dimensions, the $V_{j,k,l,p}$ entries corresponding to these orbitals will be[10]

$$V_{j,k,l,p} = \frac{1}{2}\int d^3\vec{x}\int d^3\vec{r}\frac{\psi_j^*(\vec{x}-\vec{r})\psi_k(\vec{x}-\vec{r})\psi_l^*(\vec{x})\psi_p(\vec{x})}{|\vec{r}|} \tag{19}$$

- While this may not yield an integral which is solvable by hand, it will be tractable numerically (and can be re-used any time the same basis set is used)

- Note that this doesn't tell us **how** to choose the orbitals, and in practice this will have a large effect on how well our calculation performs (mathematically speaking if we take enough orbitals, than any basis set will work, but may render the calculation infeasible given it has to run on real hardware), one natural choice is whatever basis set diagonalises the interaction free version of the Hamiltonian (i.e. solutions to the non-interacting Shcrödinger equation), but there may be better choices

- The Hamiltonians we are interested in solving therefore take the form

$$\hat{H} = \sum_j U_j \hat{c}_j^\dagger \hat{c}_j + \sum_{j,k} J_{j,k}\left(\hat{c}_j^\dagger \hat{c}_k + \hat{c}_k^\dagger \hat{c}_j\right) - \sum_{j,k,l,p} V_{j,k,l,p}\hat{c}_j^\dagger \hat{c}_l^\dagger \hat{c}_k \hat{c}_p \tag{20}$$

- We have a problem though, the interaction term means that the Hamiltonian for one particle is no longer independent from the others, in fact the diagonal basis (and therefore which modes we need to include for an accurate description) even depends on which other modes are occupied

- There are (roughly speaking) three options to solve problems involving strongly interacting electrons:

  1. Perform full diagonalisation of an $\binom{n_{\text{mode}}}{n_{\text{electron}}} \times \binom{n_{\text{mode}}}{n_{\text{electron}}}$ Hamiltonian using standard matrix methods[11], this works in principle but at least if the number of electrons grows with the number of modes, it scales exponentially so is only practical at small sizes (something like 20 orbitals with 10 electrons for a laptop and probably not more than 50 or 60 half-filled orbitals on a supercomputer).

[10] the $\frac{1}{2}$ factor is because otherwise each interaction would be double counted

[11] I have a published paper where my contribution was doing this, see https://arxiv.org/abs/1401.5680note that the standard name for this technique "exact diagonalisation" is misleading, it is actually approximate numerical diagonalisation using standard algorithms in Matlab (which in turn uses the same ARPACK and LAPACK packages as Python)

2. Make some approximations, such as Hartree-Fock which approximates as a single Slater determinant (orbitals either completely occupied or completely empty in some basis), or many, many other approximations. Density Functional Theory (DFT) is a popular choice here, where the effect of interactions from other electrons are treated as additional potentials. Lots of work on this, it can work ok sometimes but not others. Often it is relatively easy to check how good an approximation is (by checking self consistency or comparing energies if finding a ground state) even if finding an approximation is hard.
3. Use a quantum computer: even a relatively small perfect quantum computer (say 100 perfectly noise-free qubits) could solve problems that supercomputers can't, it gets messier for noisy quantum computers (often bootstrapping off of approximate classical methods), and may need more coherence to see and advantage than optimisation

- For option 3, simulating dynamics is just exponentiation of Pauli strings, and ground states could (in principle) be found by simulating adiabatic evolution, the step that really limits us here is building a good enough quantum computer